2	9							5
							2	
6		8	9					
			7				9	4
	4	3			5			6
			3	1				8
	2			5			3	
				4				9
						6	7	

Data Science and Machine Learning

SOLUTIONS

2	9	7	4	3	1	8	6	5
3	1	4	5	6	8	9	2	7
6	5	8	9	7	2	1	4	3
5	8	1	7	2	6	3	9	4
7	4	3	8	9	5	2	1	6
9	6	2	3	1	4	7	5	8
8	2	9	6	5	7	4	3	1
1	7	6	2	4	3	5	8	9
4	3	5	1	8	9	6	7	2

Module 1

Data Science

Massive data
Diversified data
Accumulated data
} are you used to manipulated

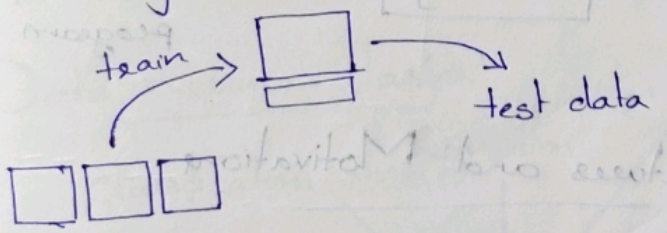
In data science data are used for

1. Prediction
2. to uncover hidden pattern
3. better decision

For analysing data we have to

1. Data gathering
2. Data preparation
3. Data cleaning
4. Data Transformation

Training Data



Artificial Intelligence

Linguistics Vision

Language Synthesis Robotics

Sensor

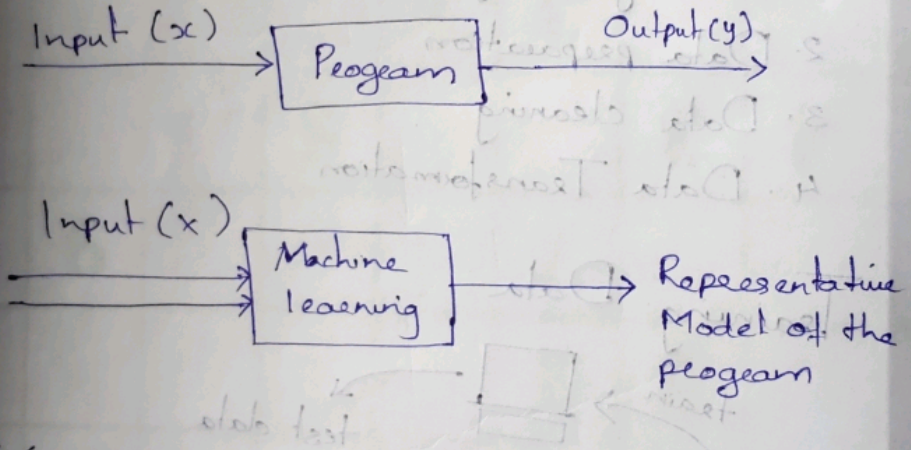
Machine learning

Support vector machines, kNN

Decision trees, Bayesian learning

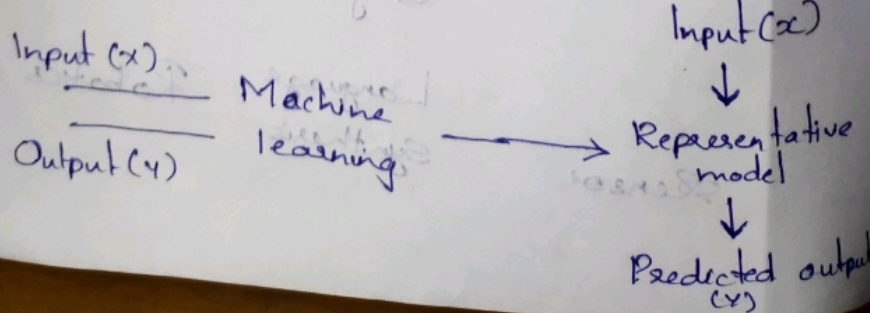
Deep

Traditional program and Machine Learning



Key Features and Motivations

1. Extracting Mean
2. Building Representative Models



3. Combination of Stats

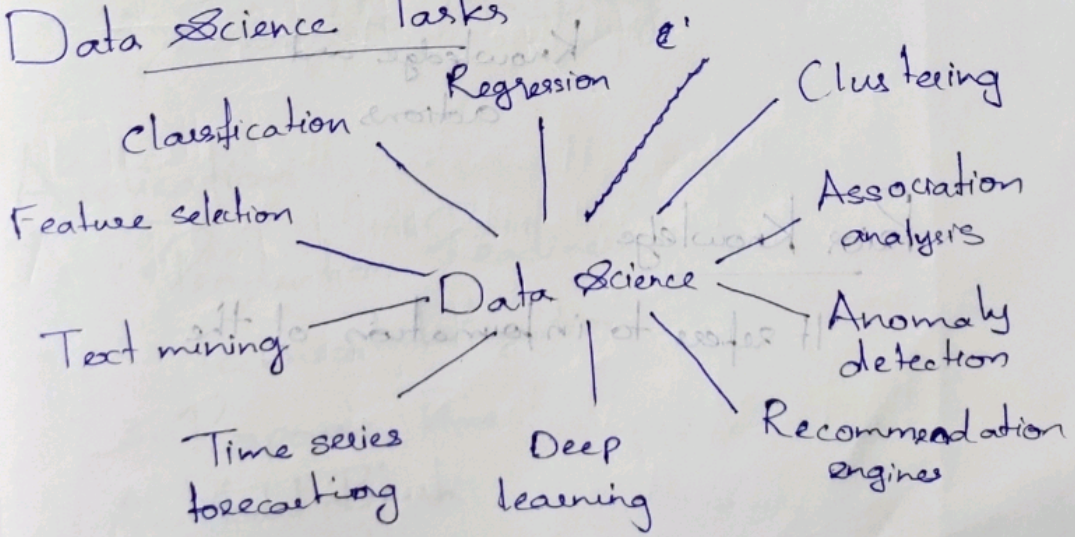
5. Associated Fields

1. Descriptive statistics
2. Exploratory visualization
3. Dimensional slicing
   - Slicing
   - Pivoting
   - Hypothesis testing
   - Data engineering
   - Business intelligence (BI)

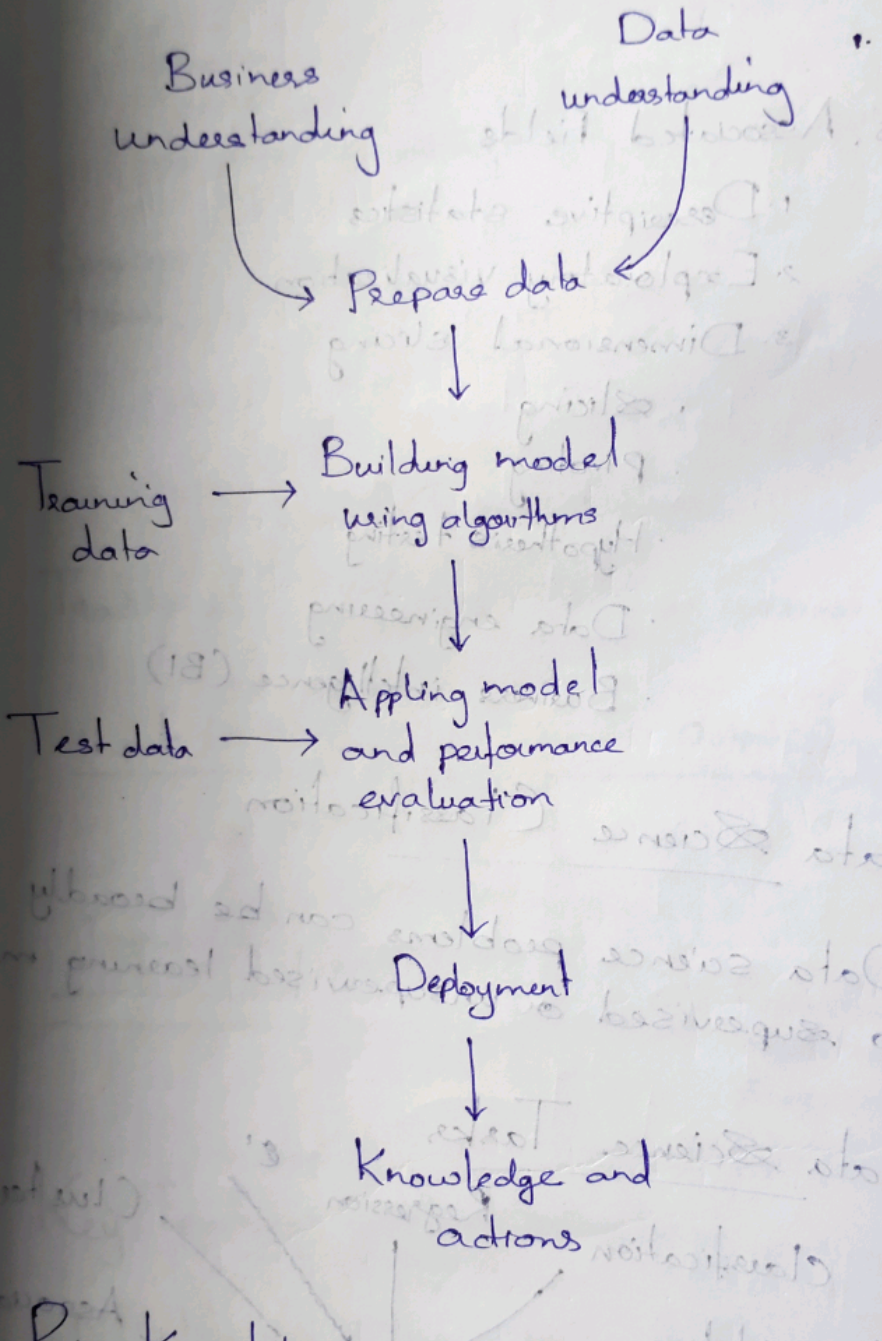
Data Science Classification

Data science problems can be broadly categorized into supervised or unsupervised learning models.

Data Science Tasks



Data Science Process



Prior Knowledge

It refers to information of the

Data Preparation

1. Data Exploration
2. Data Quality
3. Missing Values
4. Data Types and Conversion
5. Transformations
6. Outliers
7. Feature Selection
8. Data Sampling

Modelling

1. Training dataset and Testing Datasets
2. Learning Algorithms
3. Evaluation of the Model
4. Ensemble Modeling

Application

1. Production Readiness
2. Technical Integration
3. Response Time
4. Model Refresh
5. Assimilation

Knowledge

The data science process provide a framework to extract nontrivial information from data.

Data Exploration

Data Set

1. Descriptive Statistics
   - 2 types
     1. Univariate Exploration
     2. Measure of Spread

$$\text{Variance} = S^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

(x_i: value; μ: mean of the Sample)

Deviation :

$$\text{Standard deviation} : SD = \sqrt{\frac{\Sigma (x_i - \mu)}{N}}$$

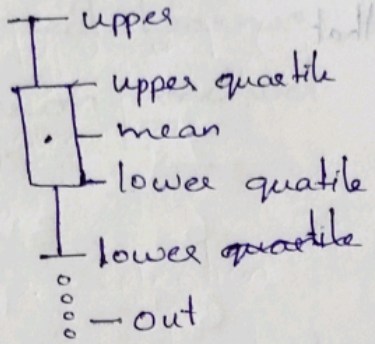
2. Multivariate Exploration

Correlation : It measures the statistical relationship between two attributes, particularly dependence of one attribute

Data Visualization

1. Univariate
   1. Histogram
   2. Quartile

Diagram shows that the quartile charts for all four attributes of the Iris dataset are plotted side by side. Petal length can be observed as having the broadest range and the sepal width has a narrow range, out of all of the four attributes



3. Distribution Chart

$$f(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{(x-\mu)^2/2\sigma^2}$$

1. Scatterplot
   1. Iris setosa
   2. Iris versicolor
   3. Iris virginica
2. Bubble Chart
3. Density Chart

Module 2

Machine Learning

The field of machine learning provides a set of algorithms that



Reality → Machine learning → face / Not face

Machine Learning in Practice

1. Data collection
2. Data Exploration and preparation
3. Model training
4. Model evaluation
5. Model improvement

Types of Input Data

1. Unit of observation
   Unit of analysis

Types of Machine Learning Algorithms

1. Predictive model

3 types of classification algorithms

1. Supervised Learnes
2. Regression method
3.
4.

1 Based on a survey conducted in an institution students are classified based on the 2 attribute of academic excellence and other activities. Given the following data identify the classification of a student with $x = 5$ and $y = 7$ using KNN algorithm (choose K as 3).

	X (Academic Excellence)	Y (Other Activities)	Z (Classification)
Student 1	8	6	Outstanding
Student 2	65	6	Good
Student 3	7	3	Good
Student 4	6	9	Outstanding
Student 5	5	7	?

$$\text{Distance}(P, Q) = \sqrt{(P_1 - Q_1)^2 + (P_2 - Q_2)^2 + \cdots + (P_k - Q_k)^2}$$

Let 'P' be student (5) with $x = 5$ and $y = 7$

'q' be students who are already classified

$$\text{distance}(P, Q) = \sqrt{(P_1 - Q_1)^2 + (P_2 - Q_2)^2}$$

dist (stud 5, stud 1) $= \sqrt{(5-8)^2 + (7-6)^2} = 3.162$

dist (stud 5, stud 2) $= \sqrt{(5-5)^2 + (7-6)^2} = 1$

dist (stud 5, stud 3) $= \sqrt{(5-7)^2 + (7-3)^2} = 4.472$

dist (stud 5, stud 4) $= \sqrt{(5-6)^2 + (7-9)^2} = 2.236$

	X	Y	Z	K = 3	
Stud 1	8	6	3.162	Yes	Outstanding
Stud 2	5	6	1	Yes	Good
Stud 3	7	3	4.472	No	X
Stud 4	6	9	2.236	Yes	Outstanding

∴ Student 5 in an outstanding student

2 Perform KNN classification algorithm on following dataset and predict the class for X (P1 = 3 and P2 = 7) K = 3.

P1	P2	Class
7	7	False
7	4	False
3	4	True
1	4	True
3	7	?

Distance $(P, Q) = \sqrt{(P_1 - Q_1)^2 + (P_2 - Q_2)^2 + \cdots + (P_n - Q_n)^2}$

Let 'P' be X with P1 = 3 and P2 = 7

dist (X, A) $= \sqrt{(3-7)^2 + (7-7)^2} = 4$

(X, B) $= \sqrt{(3-7)^2 + (7-4)^2} = 5$

(X, C) $= \sqrt{(3-3)^2 + (7-4)^2} = 3$

(X, D) $= \sqrt{(3-1)^2 + (7-4)^2} = 3.605$

	P_1	P_2	Z	K = 3	Class
A	7	7	4	Yes	False
B	7	4	5	No	False
C	3	4	3	Yes	True
D	1	4	3.605	Yes	True

Majority True (C, D)

So, X is True

Naive Bayes Algorithm

Naive Bayes Problem / Classifier Problem

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D_1	Sunny	Hot	High	Weak	No
D_2	Sunny	Hot	High	Strong	No
D_3	Overcast	Hot	High	Weak	Yes
D_4	Rain	Mild	High	Weak	Yes
D_5	Rain	Cool	Normal	Weak	Yes
D_6	~~Overcast~~ Rain	Cool	Normal	Strong	No
D_7	~~Sunny~~ Overcast	Cool	Normal	Strong	Yes
D_8	Sunny	Mild	High	Weak	No
D_9	~~Rain~~ Sunny	Cool	Normal	Weak	Yes
D_{10}	Rain	Mild	Normal	Weak	Yes
D_{11}	Sunny	Mild	Normal	Strong	Yes
D_{12}	Overcast	Mild	High	Strong	Yes
D_{13}	Overcast	Hot	Normal	Weak	Yes
D_{14}	Rain	Mild	High	Strong	No
D_{15}	Sunny	Cold	High	Strong	?

$P(\text{PlayTennis} = \text{yes}) = 9/14 = .64$

$P(\text{PlayTennis} = \text{no}) = 5/14 = .36$

Outlook	Y	N
Sunny	2/9	3/5
Overcast	4/9	0
Rain	3/9	2/5

Temperature	Y	N
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Y	N	Wind	Y	N
High	3/9	4/5	Weak	6/9	2/5
Normal	6/9	1/5	steong	3/9	3/9

(Outlook - sunny, Temperature - cool, Humidity - high, Wind - steong)

V_{NB} = argmax $P(v_j) \prod_i P(a_i/v_j)$
$v_j \in \{yes, no\}$

= argmax $P(v_j)$ P(Outlook - sunny $|v_i$) P(Temperature - cool $|v_j$)
$v_j \in \{yes, no\}$

P(Humidity - high $|v_j$) P(Wind - steong $|v_j$)

V_{NB}(Yes) = P(Yes) P(Sunny|Yes) P(cool|yes) P(high|Yes) P(steong|Yes = .0053

V_{NB}(no) = P(no) P(sunny|no) P(cool|no) P(high|no) P(steong|no) = .0206

.64 (2÷9) × (3÷9) × (3÷9) × (3÷9)

$P(yes) = \frac{.0053}{(.0053 + .0206)} = \frac{.0053}{0.0259} = 0.2046 = \underline{0.20}$

$P(no) = \frac{.0206}{(.0206 + .0053)} = \frac{.0206}{0.0259} = \underline{0.81}$

The probability of the person playing tennis is classified as 'no' as the probability of outcome no for the given condition outlook-sun, humidity =

high wind : strong) is higher the probability of outcome yes for the same condition.

Example 2

Estimate conditional probabilities of each attributes {color, legs, height, smelly} for the species classes: {M, H} using the data given

Language

No	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H

6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

New Instance

(Color = Green, legs = 2, Height = Tall, and Smelly = No)

$P(M) = \frac{4}{8} = 0.5$ $P(H) = \frac{4}{8} = 0.5$

Color	M	H
White	2/4	3/4
Green	2/4	1/4

Legs	M	H
2	1/4	4/4
3	3/4	0/4

Height	M	H
Tall	3/4	2/4
Short	1/4	2/4

Smelly	M	H
Yes	3/4	1/4
No	1/4	3/4

5	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

New Instance

(Color: Green, Legs: 2, Height: Tall, and Smelly: ?)

$P(M) = \frac{4}{8} = 0.5$ $P(H) = \frac{4}{8} = 0.5$

Color	M	H
White	2/4	3/4
Green	2/4	1/4

Legs	M	H
2	1/4	N/A
3	3/4	0/4

Height	M	H
Tall	3/4	2/4
Short	1/4	2/4

Smelly	M	H
Yes	3/4	1/4
No	1/4	3/4